DOI: 10.1111/mice.12712

INDUSTRIAL APPLICATION

COMPUTER-AIDED CIVIL AND INFRASTRUCTURE ENGINEERING WILEY

Vortex-induced vibration measurement of a long-span suspension bridge through noncontact sensing strategies

Jian Zhang[1,2] | **Liming Zhou**[3] | **Yongding Tian**[2,4] | **Shanshan Yu**[2] | **Wenju Zhao**[5] | **Yuyao Cheng**[6]

[1] Jiangsu Key Laboratory of Engineering Mechanics, Southeast University, Nanjing, China

[2] School of Civil Engineering, Southeast University, Nanjing, China

[3] Structural Health Monitoring and Control Institute, Shijiazhuang Tiedao University, Shijiazhuang, China

[4] Department of Civil Engineering, Monash University, Melbourne, Australia

[5] College of Civil Engineering and Architecture, Henan University of Technology, Zhengzhou, China

[6] Faculty of Civil Engineering and Mechanics, Jiangsu University, Zhenjiang, China

Correspondence
J. Zhang, School of Civil Engineering, Southeast University, Nanjing, 210096, China.
Email: jian@seu.edu.cn

Funding information
National Key R&D Program of China, Grant/Award Number: 2018YFC0705601; National Science Foundation of China, Grant/Award Number: 51778134; Research project of Guangdong Provincial Communication Group Company Limited, Grant/Award Number: 2020103.1; Key R&D Project of Hebei Province, Grant/Award Number: 19210804D; Science and Technology Innovation Project of China Energy Investment Corporation, Grant/Award Number: SHGF-15-41

Abstract
Long-span bridges are susceptible to vortex-induced vibration (VIV), which affects the serviceability and safety of bridges when the vibration amplitude is too large and lasts for a long time. Traditional contact-type sensing technologies (i.e., accelerometers and linear variable differential transformer) are inconvenient and dangerous to be installed on long-span bridges for monitoring VIV events. To address this limitation, this article focuses on the VIV measurement of a long-span suspension bridge through noncontact sensing strategies. The contribution of this article lies in (1) noncontact sensing technologies including microwave radar, optical camera and video equipment were employed to measure multiple-point displacements of the studied bridge under VIV events; (2) dynamic properties of the bridge (i.e., natural frequency, damping ratio, mode shapes) and characteristics of the VIV event (i.e., single-mode vibration and dominant vibration mode switch) were identified by analyzing monitoring data; (3) an early warning framework for VIV event of long-span suspension bridges was proposed based on monitored dynamic responses and wind fields; specifically, two indicators, the dominant vibration frequency and the similarity between bridge shape and vibration mode shape, were proposed to identify the VIV event, and then the root mean square (RMS) of measured response was further calculated to determine whether there is a need to trigger the warning system or not. The proposed noncontact VIV measurement strategy has the advantage of rapid measurement of vibration magnitude, rapid identification of dynamic properties of the studied bridge and characteristics of the VIV event, which are helpful for the government and bridge owners to make decisions on vibration mitigation measures and to avoid safety issues.

1 | INTRODUCTION

Bridges are the most important civil infrastructures in road networks, and reliable monitoring techniques are urgently needed to ensure their safety. Image-based techniques with computer vision and deep-learning techniques have been employed for surface cracks/defects detection of bridges recently (Busca et al., 2014; Ni, Zhang &

Noorimohammad, 2019; Ni, Zhang & Chen, 2019; Yoon et al., 2020). Long-span bridges have been constructed around the world for spanning seas, rivers, and valleys; however, they are sensitive to wind loads. For long-span bridges, wind excitations may cause large structural vibrations, which involve the aero-elastic phenomena of buffeting, flutter, and vortex-induced vibration (VIV) and are often critical in the safety and serviceability of long-span bridges (Scanlan, 1981). Structural health monitoring (SHM) systems have been installed on bridges to monitor structural responses and operational conditions (Brownjohn et al., 2011; Chen et al., 2017; Sohn et al., 2004; Lynch, 2007; Ou & Li, 2010). Based on long-term recorded SHM data of wind and wind-induced structural response, many studies have been conducted in the past decades to explore wind characteristics and their effects (Hao & Wu, 2018; Kim & Adeli., 2005; Li, et al., 2011; Nagai et al., 2004; Powell et al., 2003; Wang & Adeli, 2015; Zhou & Sun, 2018). The buffeting analysis of long-span bridges considering time-varying wind coherence was proposed and applied to the Stonecutters Bridge (Tao et al., 2020). Flutter is the most dangerous type of wind-induced vibrations for long-span bridges and was the basic reason causing the collapse disaster of the old Tacoma Narrow Bridge in 1940 (Xiang et al., 2005; L. Zhu et al., 2013; L. D. Zhu et al., 2020). VIV event is prone to occur on long-span bridges at low wind speed because of the lower and denser natural frequencies, which will adversely affect bridge health and driving safety when the vibration amplitude is large and lasts for a long time. VIV is the vibration caused by a fluid flow that produces vortex shedding at or near a structural natural frequency of the bridge. Long-span structures, such as suspension bridge, cable-stayed bridge, and box-girder bridge, are prone to be affected by VIV event. For example, a VIV phenomenon was observed on a long-span suspension bridge, the Great Belt East Bridge, after this event, guide vanes were designed and implemented to mitigate the vibration amplitude of the this bridge (Frandson et al., 2001). The VIV event occurred on a ten-span continuous steel box-girder bridge, Trans-Tokyo Bay Crossing Bridge; during this event, the wind speed was approximately 16–17 m/s and the wind direction was about ±20° normal to the bridge axis, and a tuned mass damper was installed on this bridge to mitigate the vibration (Fujino & Yoshida, 2002). The VIV usually cannot produce destructive damages to bridges, but it can induce large vibration amplitudes, which can make drivers/pedestrians feel uncomfortable and fatigue; when the VIV event lasts for a long time, it can result damage. Therefore, it is very important to measure the vibration amplitude of VIV and identify dynamic properties of a bridge.

Most SHM systems have installed contact-type sensing technologies such as electrical resistance strain sensors, vibrating wire strain gauges, and piezoelectric accelerometers, and they provide excellent opportunities to investigate the VIV mechanism and early warning of long-span bridges. Li et al. (2014) investigated the wind conditions and the mechanism of VIV of Xihoumen suspension bridge based on long-term field monitoring data and concluded that the vortex shedding will gradually strengthen and extend to the whole lower surface of downstream deck and the tail of upstream deck from the development stage to the lock-in stage. Hwang et al. (2019) investigated the cause of VIV event of a suspension bridge, Yi Sun-Shin Bridge, and stated that the vibration amplitude of bridge under VIV event was sensitive to the Scruton number. They pointed out two possible reasons for VIV event, one was that the vibration mode with low damping ratio may be triggered as the dominant vibration mode, another reason was that the VIV-triggered vibration mode of long-span bridges could happen at those vibration modes with low damping ratios. Macdonald et al. (2003) investigated the VIV event of a cable-stayed bridge, Second Severn Bridge, although the wind tunnel test was performed in the design phase and some mitigation measures were implemented on this bridge, VIVs were still being monitored by the installed SHM system, and then they further installed baffle plates under the bridge to mitigate the large vibration amplitude of the bridge. Although SHM systems have the capability of long-term monitoring and early warning of VIV event, the whole system is expensive and not all bridges in the road network are installed with various sensors. In addition, when VIV event is occurring, installing sensors on a bridge is dangerous and time-consuming. Therefore, new measurement techniques having the capability of rapid measurement and low expense are promising techniques for health monitoring and early warning of bridges.

In recent years, noncontact measurement technologies have been rapidly developed, which are ideal measurement techniques for coping with emergency events (i.e., vortex-induced vibration, post-disaster inspection). Existing noncontact measurement techniques include LiDAR technology (Nassif et al., 2005; Oh et al., 2017; H. S. Park et al., 2007; S. W. Park et al., 2015), microwave radar (Amies et al., 2019; G. Zhang et al., 2020; Zhao et al., 2020), optical camera sensing (Busca et al., 2014; Tian et al., 2020; Xu et al., 2018; Yu et al., 2020; B. Zhang & Zhang, 2020), and global position system (GPS) (Xu & Chan, 2009). Microwave radar and optical camera have the capability of measuring multipoint displacements of bridges from a long distance, and they are immunity to electromagnetic interference and ease of installation. For example, Pieraccini et al. (2019) used the microwave interferometry radar to monitor displacements of a railway bridge for assessing the dynamic impact factor. Yu et al. (2019) developed a new

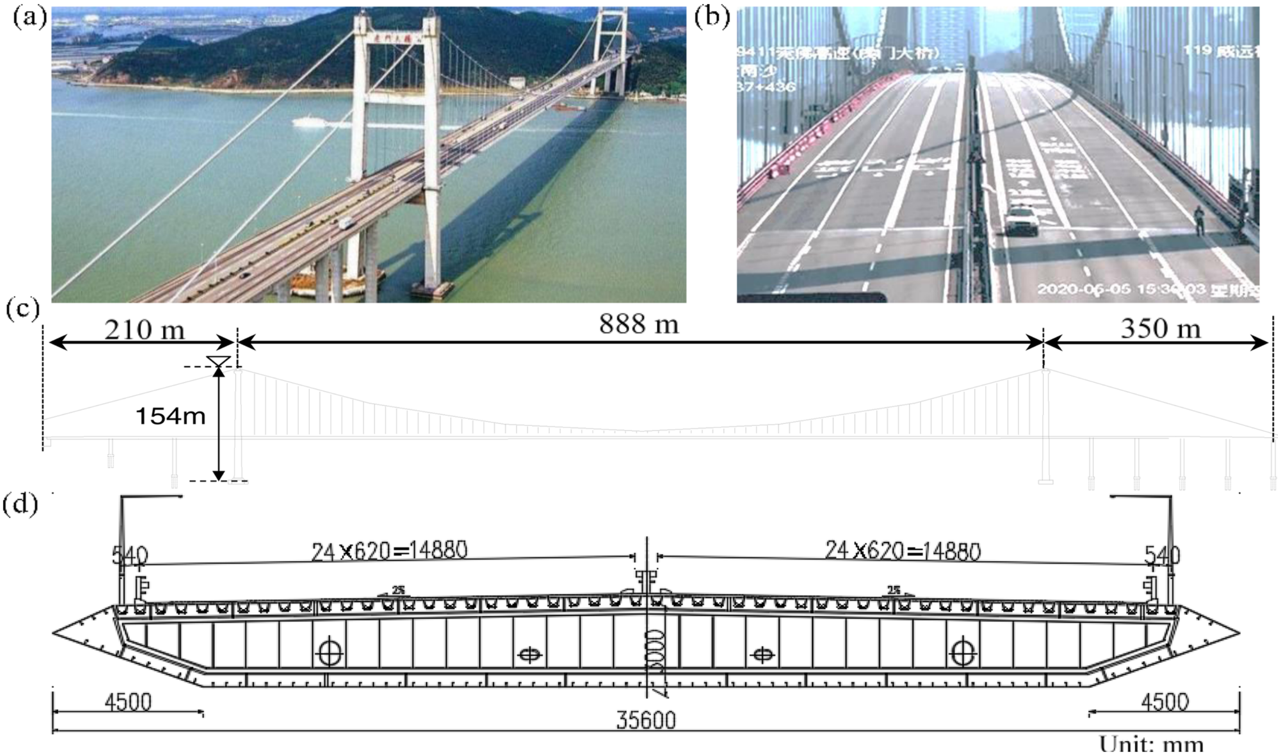

FIGURE 1 Photograph and span arrangement of a long-span suspension bridge: (a) panorama view; (b) vortex-induced vibration; (c) elevation view; (d) cross-section view of the bridge deck

vision-based method for measuring the mid-span deflection and multiple cable tensile force of Sutong Yangtze River bridge. Ribeiro et al. (2014) developed a video-based noncontact measurement system and applied this system to measure the dynamic displacement of a railway bridge under a moving train. However, to the authors' knowledge, there is no existing study related to noncontact vibration measurement of bridges when VIV event occurs.

The objective of this study is to use noncontact measurement devices for measuring the vibration amplitude and dynamic properties of a long-span suspension bridge under VIV event. An early warning framework for VIV event was proposed based on monitored multiple-point displacements and identified dynamic properties. The structure of this paper is as follows: Overview of field testing of a long-span suspension bridge is introduced in Section 2. Section 3 investigates dynamic properties of the bridge including the data processing procedures, results of natural frequencies, and damping ratios; in Section 4, characteristics of VIV event including single-mode vibration and dominant vibration mode switch between adjacent modes are analyzed by using noncontact measurements, then the statistic relationship between dynamic responses and wind field is investigated. An early warning framework for VIV events is proposed and demonstrated; finally, conclusions are given in Section 5.

2 | OVERVIEW OF FIELD TESTING

The investigated bridge in this study will be first described in this section, and then noncontact measurement scheme for VIV monitoring of this bridge is described. After that, developed noncontact displacement measurement devices including microwave radar, optical camera system, and video device are introduced in detail; finally, preliminary results of this bridge will be presented by analyzing measured dynamic responses.

2.1 | Bridge description

The studied bridge is a sea-crossing long-span suspension bridge located at Guangzhou province, the south part of China. This bridge was started to construct in 1992 and opened to traffic in 1997. The photograph and span arrangement of this bridge is shown in Figure 1. It has a main span of 888 m, side span of 210 m and 350 m, respectively. The height of the bridge tower is 154 m and the width of the bridge deck is 35.6 m (Figure 1d). An unexpected VIV event was observed on this bridge on May 5, 2020, and the two-way traffic control measures were immediately taken to avoid safety issues. The VIV was possibly caused by the temporary cover placed on bridge deck during the pro-

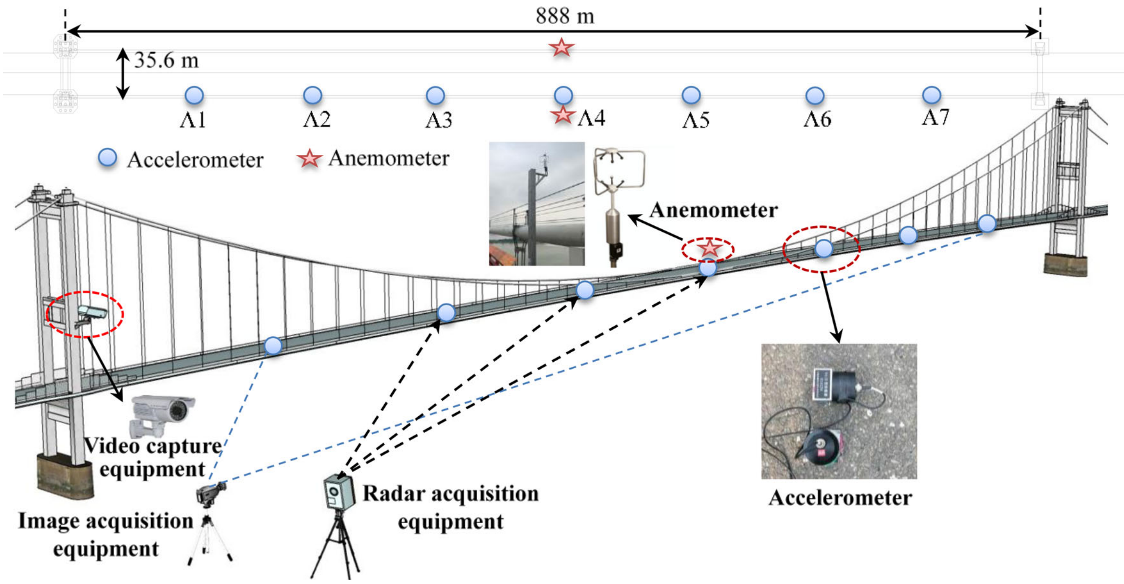

FIGURE 2 Sensors arrangement of the studied bridge

cess of vertical hanger replacement. It changed the aerodynamic shape of the steel box girder, resulting in dramatic oscillations under low wind speeds. To get a better understanding of the characteristics of the VIV event, dynamic responses of the bridge under VIV event were measured by multiple noncontact measurement devices, including microwave radar, optical camera, and video equipment. During testing period, the bridge was not opened to public traffic. The noncontact measurement scheme will be introduced in the following section in detail.

2.2 | Noncontact measurement scheme

Traditional contact-type sensing technologies are dangerous and time-consuming to be employed on this bridge when VIV event occurs. Therefore, a noncontact measurement scheme (Figure 2) is proposed in this article to monitor dynamic responses of the bridge from a long distance. Noncontact measurement devices including microwave radar, optical camera, and video device were adopted to measure multiple point displacements of the bridge under VIV event. Microwave radar and optical camera were installed under the bridge to monitor displacements of seven cross-section of the bridge, the video device was installed on the bridge tower to monitor the displacement of the bridge deck. To verify the correctness of the noncontact measurement device for VIV identification, accelerometers were also installed on the bridge deck for measuring dynamic acceleration responses. It should be mentioned that the distance between two adjacent accelerometers is equal to 111 m. In addition, two anemometers were installed on the upstream and downstream sides of the mid-span section of the bridge for monitoring surrounding wind loads.

2.3 | Noncontact measurement device

2.3.1 | Microwave radar

The developed microwave radar system for noncontact displacement measurement is briefly presented in this section, more details about this device can be found in our previous research work (G. Zhang et al., 2020). By transmitting and receiving microwaves, the developed equipment has the capability of measuring 0.01 mm displacement with the device-to-bridge distance of 1 km. In particular, 16 single-shot antennas were integrated with the device, and the field angle of the developed equipment's antenna can up to 10° (horizontal) × 10° (vertical). Furthermore, auxiliary units such as electronic compass, laser pointer, and laser range finder, were also equipped on this device for conveniently measuring information of the energy center point.

2.3.2 | Optical camera

In field testing, a camera with resolution of 2048 × 2048 pixel, lens with focal length of 35 mm and sampling rate of 4 Hz were adopted to capture vibration images of the bridge. The pitching angle of the camera is estimated as 7°. The farthest target point was about 840 m away from the measurement system. After image sequences of the bridge were measured, a template matching algorithm was employed to extract dynamic displacements. The principle of this algorithm can be described as follows: A rectangular region of interest (ROI) needs to be chosen in the reference image first, then the digital image correlation method (DICM) based on inverse-compositional Gauss–Newton (IC-GN) algorithm (Baker & Matthews, 2004) is adopted to determine sub-pixel displacement of the ROI by optimizing the following zero-mean normalized sum of squared difference (ZNSSD) function:

$$C_{ZNSSD} = \sum_{y=-M}^{M} \sum_{x=-M}^{M} \left[\frac{f\left(W\left(x, y; \Delta p\right)\right) - f_m}{f_s} - \frac{g\left(W\left(x, y; \Delta p\right)\right) - g_m}{g_s} \right]^2 \tag{1}$$

where f and g denote the grayscale level at the point (x, y), f_m and g_m are the mean intensity value of the subset, M is the width of the subset and W(x, y; Δp) is the warp function. Based on the existing research, it is generally believed that the larger the template size, the higher the level of sub-pixel accuracy, but the lower the computational speed. Therefore, it is necessary to balance comprehensively according to the actual situation. Taking this study as an example, seven regions of interest are selected, the template size is 21, the step size is 5, the Zero-mean normalized cross-correlation (ZNCC) threshold is 0.8, the maximum number of iterations is 20, the search radius is set as 20, and the post-processing frequency is about 8 Hz. Therefore, it takes about 5 minutes to process all the images of 10 minutes. Finally, the pixel displacement is converted into the physical displacement through a pixel-to-millimeter coefficient based on the object distance and the pitching angle (Pan & Tian et al., 2016). DICM is slower than other matching algorithms, but its robustness and accuracy are more reliable, and it can be qualified for structures whose features are not significant at a long distance. In addition, the displacement measurement errors of target points with different object distances are not consistent. In contrast, the image signal processing time is more time consuming than the traditional method, but the equipment erection of image sensor is time-saving, and it is much more suitable for urgent tasks.

2.3.3 | Video device

A video camera was installed on the bridge tower for recording vibration images of the bridge deck. The installation of this video camera is shown in Figure 2. The video recording equipment adopts infrared high-definition spherical camera (DH-SD-6A9230F-HNI), which has the characteristics of clear image, digital, intelligent, and convenient installation. The rotation speed of pan-tilt-zoom (PTZ) camera can be automatically adjusted according to the zoom ratio of lens. The camera has a resolution of 1024 × 1024 pixel and sampling rate of 24 Hz.

2.3.4 | Accelerometer and anemometer

In addition to noncontact measurement devices, seven accelerometers were installed on one side of the bridge deck for measuring dynamic accelerations of the bridge under VIV event (Figure 2). The accelerometer with model of 941B ultra-low frequency vibration meters was adopted in field testing. It is a multi-functional instrument for ultra-low frequency or low-frequency vibration measurement and mainly used for tremor measurement. This accelerometer can collect vibration responses of the bridge in both vertical and transverse direction. It has the capability of measuring accelerations up to 20 m/s^2, and the measurement resolution and sensitivity are equal to 5×10^{-6} m/s^2 and 0.3 V·s/m. Accelerometers were installed on every 1/8 section of the bridge and the sampling rate was set to 50 Hz. In addition, two ultrasonic anemometers with model of 81000 were installed on the middle span for measuring three-dimensional wind speed and wind direction. Lower output rate of 10 Hz was chosen in field testing because lower sampling rate is more immune to outliers in the samples.

2.4 | Preliminary results

Measured displacements of the bridge by noncontact measurement device are first described and the correctness of measured displacements is verified by comparing the results obtained from microwave radar, optical camera, and video device, then the statistics analysis between dynamic responses and wind fields are analyzed and dynamic properties of this bridge will also be presented in this section.

2.4.1 | Measured displacements

To verify the reliability of various monitoring equipment for displacement monitoring, measured dynamic displacements at 11:54 am on May 8, 2020, were selected for comparison in time domain and frequency domain as shown in Figure 3. The upper part of Figure 3 shows the photograph of microwave radar, optical camera, and video device for displacement measurement. After raw data were measured, data processing algorithms were employed to obtain dynamic displacements of the bridge. The middle part of Figure 3 shows the vertical displacement of the bridge deck

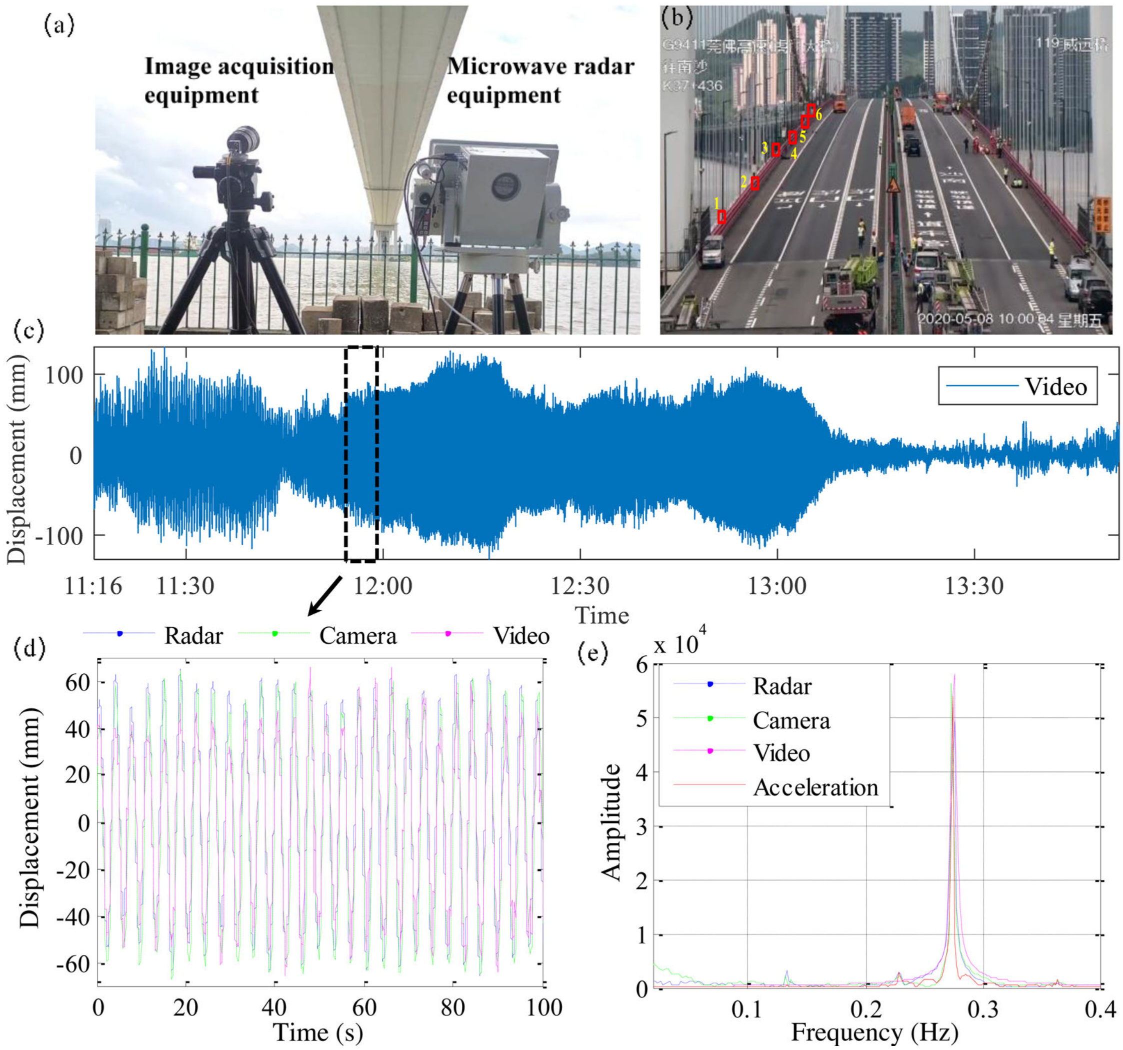

FIGURE 3 Preliminary comparison of measurement results: (a) photograph of camera and microwave radar for displacement measurement; (b) video-captured bridge vibration image; (c) extracted displacement at A1 location of the bridge; (d) displacement comparison with three noncontact measurement devices; (e) Fourier spectrum comparison of measured displacements

at 1/8 section measured by video equipment. The measured displacement by video camera starting from 11:54 am on May 8, 2020, was compared with the results obtained by microwave radar and optical camera. It can be seen that the measured displacements agree well with each other in time domain, demonstrating the correctness of measured displacements. Furthermore, Fourier spectrum of measured displacements were compared to that obtained by measured accelerations. It shows that four kinds of dynamic responses have the same frequency component and the dominant frequency of this bridge during this time is 0.273 Hz, corresponding to the fourth vibration mode of the bridge. This frequency characteristic demonstrates that a VIV event is occurring on this bridge, which will be introduced in the following sections.

2.4.2 | Measured accelerations and wind conditions

As mentioned in Section 2.3, in addition to noncontact measurement device, contact-type accelerometers were also installed on the bridge deck for investigating dynamic properties of the studied bridge. The acceleration time history at A1 location from May 8 to 14, 2020. During this period, VIV events occurred frequently on the studied bridge, and some vibration mitigation measures were adopted to minimize the vibration amplitude, such as installing spoiler and water tank. After mitigation measures were taken, the vibration amplitude of the bridge was greatly reduced and opened to traffic on May 15. It is noted that the VIV event occurred frequently from May

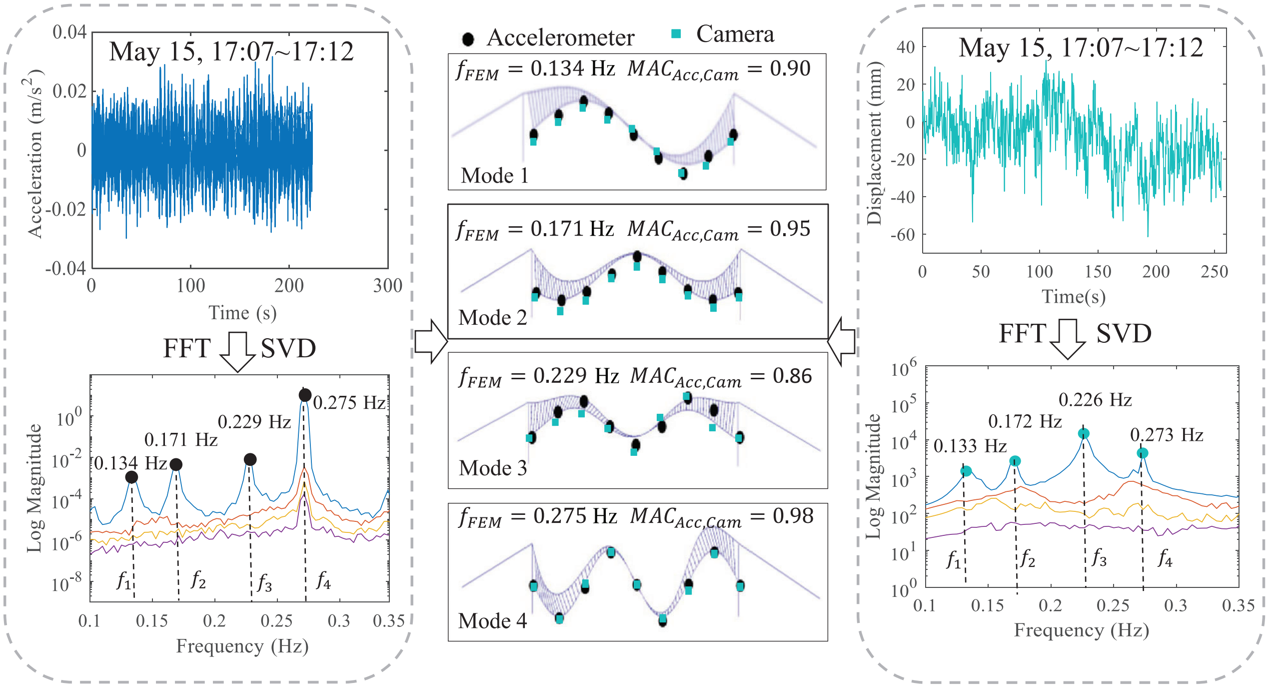

FIGURE 4 Dynamic properties identification from measured multiple-point responses: (a) dynamic accelerations; (b) comparison of identified mode shapes with calculated values from finite element model (FEM); (c) dynamic displacements

8 to 9, and the happening frequency of VIV was significantly reduced after taking mitigation measures. The statistical relationship between root mean square (RMS) of acceleration and wind fields is investigated. In this investigation, the measured data was divided into 343 segments with segment length of 30 minutes. The VIV event was separated from measured signal by performing singular value decomposition (SVD) method on multiple point acceleration responses. If the SVD spectrum has the characteristics of single peak, the corresponding data segment can be regarded as vibration signal of the bridge under VIV event. It should be noted that the normal vibration here refers to the condition that the no VIV events happen. It can be easily differentiated from the SVD spectrum with multiple peaks. It is shown that RMS of acceleration under VIV event has a large value than that under normal vibration and wind direction is mainly distributed in the range of 290–330° when VIV event occurs.

3 | DYNAMIC PROPERTIES IDENTIFICATION

Dynamic properties of the studied bridge are identified in this section for investigating the dynamic performances of the bridge under VIV event and normal vibration. Specifically, natural frequencies and corresponding modal damping ratios of the studied bridge are identified, and the relationship between modal damping rations and wind fields is also investigated.

3.1 | Data processing procedure

Dynamic properties of the bridge without VIV events are investigated by processing multiple point displacements of the bridge measured by the optical camera on May 15 and compared with those identified from measured accelerations. Identified natural frequencies and mode shapes of the bridge are shown in Figure 4. Dynamic displacements of bridge without VIV events were chosen for parameter identification. Modal parameters of this bridge were identified by using complex mode indicator function (CMIF) algorithm (Catbas et al., 2004). In this algorithm, unscaled frequency response function (FRF) is first obtained by performing cross-correlation calculation and Fourier transform on measured multiple point displacements, and then natural frequencies and mode shapes of the bridge can be extracted by using SVD method. The first four natural frequencies obtained from displacement measurements by optical camera are 0.13, 0.17, 0.23, and 0.27 Hz, respectively, which are close to those identified from measured acceleration data and calculated value from finite element model (FEM). In addition, the modal assurance criterion (MAC) value between identified mode shapes from acceleration data and displacement data are approximating to

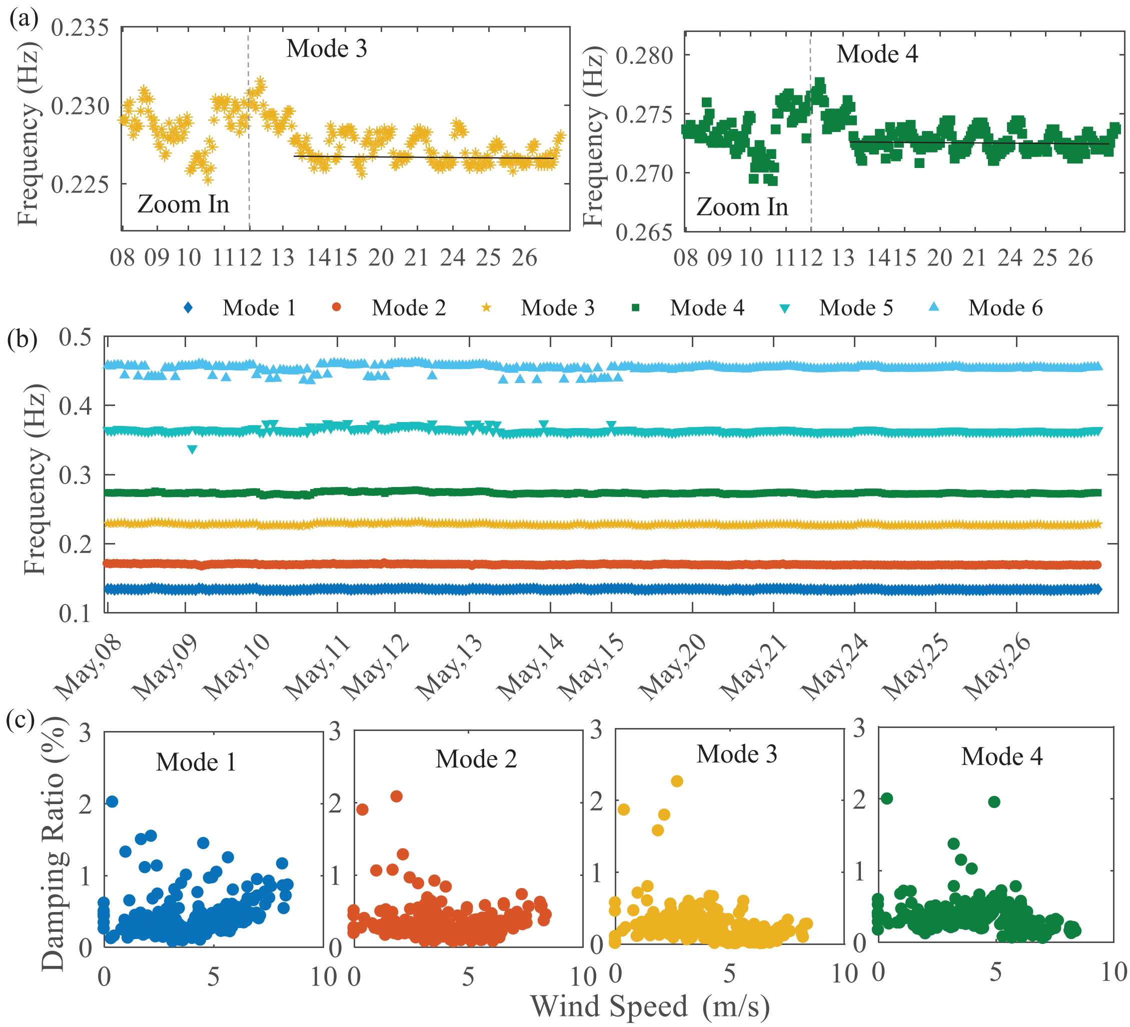

FIGURE 5 Identified natural frequencies of the studied bridge: (a) natural frequency at the third and fourth mode; (b) the first six modal frequencies; (c) the relationship between damping ratios and wind speed

0.9, demonstrating the correctness of identified dynamic properties.

3.2 | Natural frequencies

In addition to the dynamic properties of the bridge identified from noncontact measurements, statistical analysis of natural frequencies of the bridge was further investigated by using measured accelerations from May 8 to 26. Accelerometer deployment scheme is shown in Figure 2. A total of seven accelerometers were installed on the bridge deck with an equal distance of 111 m. The sampling rate is set as 50 Hz for collecting dynamic responses of the bridge under VIV event and normal vibration. A frequency domain decomposition algorithm was employed to analyze the measured dynamic responses and the identified first six natural frequencies of the bridge are shown Figure 5b. It is shown that the third and fourth natural frequencies gradually decreased from May 12 and then fluctuated around a constant value (Figure 5a). It is probably caused by the adopted vortex mitigation measures, especially the added water tank on the bridge.

3.3 | Damping ratios

Measured acceleration time histories of the bridge under normal vibration conditions were further processed for identifying damping ratios. The modal responses around each frequency component are computed by using inverse Fourier transformation, and they are further adopted to obtain free-decay responses of the bridge by using correlation function in time domain; finally, modal damping ratios of the bridge is identified by fitting the envelop data extracted from free-decay responses.

The relationship between damping ratio and wind speed is investigated. Measured dynamic accelerations from May

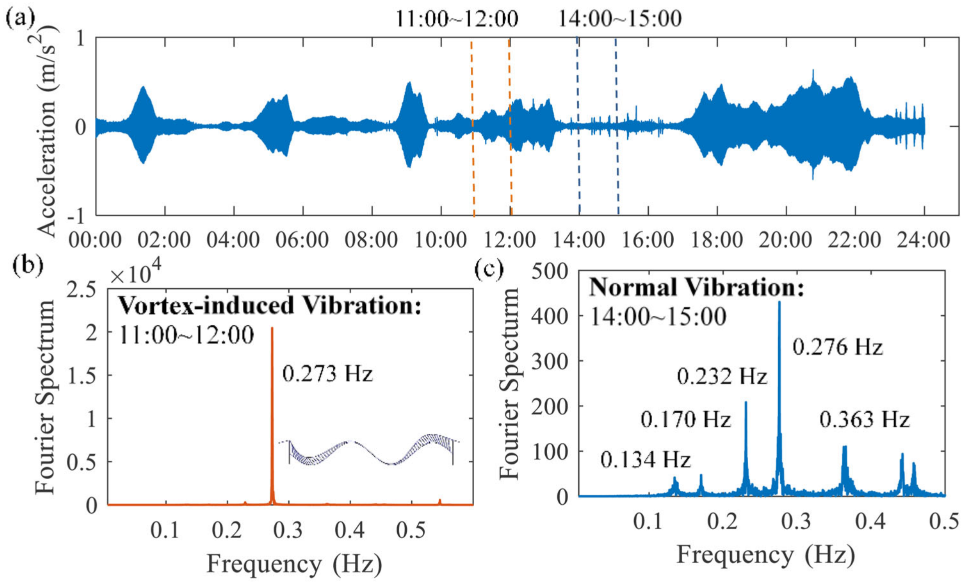

FIGURE 6 The difference of frequency characteristics between normal vibration and vortex-induced vibration (VIV) event: (a) measured acceleration on May 8; (b) Fourier spectrum of acceleration under VIV events; (c) Fourier spectrum of acceleration without VIV events

9 to May 26 were divided into 294 data segments with one-hour length data for identifying damping ratios of the studied bridge. The averaged value of wind speed and wind direction within one-hour time period was also computed. The relationship between the first four damping ratios of the studied bridge and averaged wind fields is shown in Figure 5c. It is seen that the first and second modal damping ratio slightly increases with the wind speed, whereas the third and fourth modal damping ratio does not have an obvious increasing or decreasing trend. This is reasonable because the variation of wind speed does not change too much during monitoring period (less than 9 m/s), and hence the aerodynamic damping has less effect on the total damping ratio.

4 | VORTEX-INDUCED VIBRATION IDENTIFICATION

In this section, the VIV-triggered vibration mode, the phenomenon of dominant vibration mode switch of the studied bridge and the relationship between dynamic responses and wind fields will be investigated. An early warning framework for VIV events will be proposed and demonstrated as well.

4.1 | Determination of VIV-triggered vibration mode

The difference of vibration signal between normal vibration and VIV is investigated by analyzing measured dynamic responses of the bridge. Measured acceleration responses and corresponding Fourier spectrum of the bridge on May 8 are shown in Figure 6. It is seen that the vibration signal of the bridge in normal condition is different from that under VIV event. First, the vibration amplitude of the bridge under normal condition (14:00–15:00) is much smaller than that under VIV event (11:00–12:00). Second, vibration signal of the bridge under normal condition is a kind of random vibration; however, vibration signal under VIV event is a simple harmonic vibration signal. Finally, Fourier spectrum of normal vibration signal has multiple frequency components, whereas only a single frequency component appears in the Fourier spectrum of measured signal under VIV event. Hence, we can easily distinguish VIV event from measured data by performing fast Fourier transform analysis. By analyzing measured data between 11:00 am and 12:00 am, the VIV-triggered vibration mode happened at the fourth vertical asymmetric mode with natural frequency of 0.273 Hz. The normal vibration signal between 14:00 and 15:00 was also analyzed, we can see that the corresponding Fourier spectrum has multiple frequency components.

Other than the analysis of frequency components contained in the measured signal, the bridge shape in time domain was also calculated to further determine the VIV-triggered vibration mode. An optical camera with resolution of 1024 × 1024 pixel was adopted to measure dynamic displacements of the bridge under VIV event (Figure 7a). Field of view (FOV) of the camera is approximately 700 m long, covering most region of the middle-span of the studied bridge. Dynamic displacements at seven locations (labeled as A1–A7) were extracted by using digital image

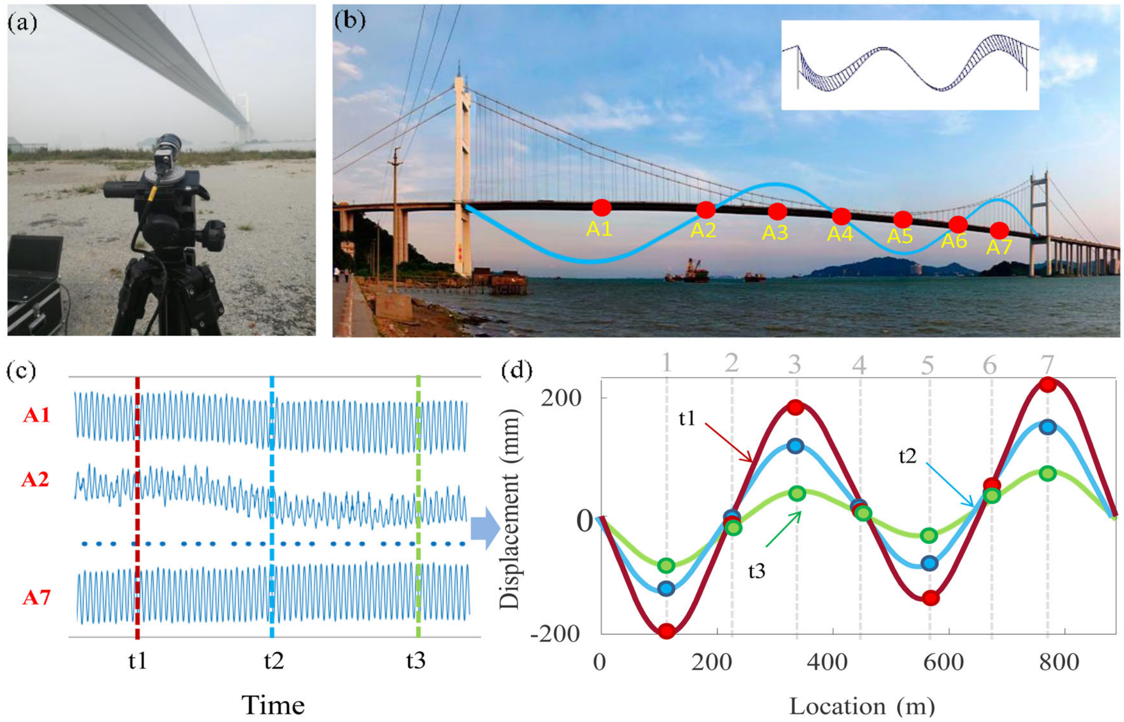

FIGURE 7 Bridge shape extraction from measured multiple point displacements by optical camera: (a) optical camera; (b) demonstration of the similarity between bridge shape and the fourth vibration mode shape; (c) measured displacements at seven locations; (d) bridge shape at different time instants

correlation algorithms as described in Section 2.3.2. Then bridge shape is obtained by selecting the displacement magnitudes of all seven locations at a specific time (Figure 7c). The bridge shapes at time T_1, T_2, and T_3 are presented in Figure 7d. It is clear that bridge shapes at those time instants have similar pattern and are the same as the fourth vibration mode shape of the bridge, demonstrating that the energy concentrating on the fourth mode when VIV event occurs. It further illustrates the single-mode properties of VIV event.

4.2 | Phenomenon of vibration mode switch

Other than the single-mode vibration characteristics of VIV, the phenomenon of vibration mode switch of the studied bridge under VIV event was also observed during field testing. This phenomenon will be investigated in this section by analyzing monitored dynamic displacements and accelerations with time-frequency spectrum method. Through the frequency spectrum analysis and the similarity analysis of extracted bridge shape of studied bridge, the VIV-triggered vibration frequencies are found to be at the second (0.171 Hz), third (0.231 Hz), and fourth (0.273 Hz) mode of the bridge, and VIV event happens frequently on the third and fourth mode. The time-frequency analysis method was further employed to analyze the measured dynamic displacements from captured videos. Captured video from 10:00 to 16:00 on May 9 were adopted to extract dynamic displacements of the bridge (Figure 8a). Time-frequency spectrum of extracted displacements is shown in Figure 8b, it clearly shows that the spectrum energy is concentrated on the third (0.231 Hz) and fourth (0.273 Hz) mode, and the corresponding vibration mode shape is shown in Figure 8c for better understanding. An important finding is that the spectrum energy distribution is switching with time. Specifically, the energy is concentrated on the third mode at 12:00 pm, whereas the energy is switched to the fourth mode at 13:00. This finding illustrates that the VIV event has the characteristics of vibration mode switching.

Furthermore, acceleration responses of the bridge are also analyzed to confirm the phenomenon of vibration mode switch identified from measured displacements. The acceleration time history at A1 location of the bridge measured on May 9, 2020, is used to calculate the time-frequency power spectrum as shown in Figure 9, it is seen from the two-dimensional diagram that the vibration mode switch phenomenon occurs at 0.231 Hz and 0.273 Hz as well (Figure 9b). The three-dimensional spectrum energy graph (Figure 9c) clearly illustrates the characteristics of

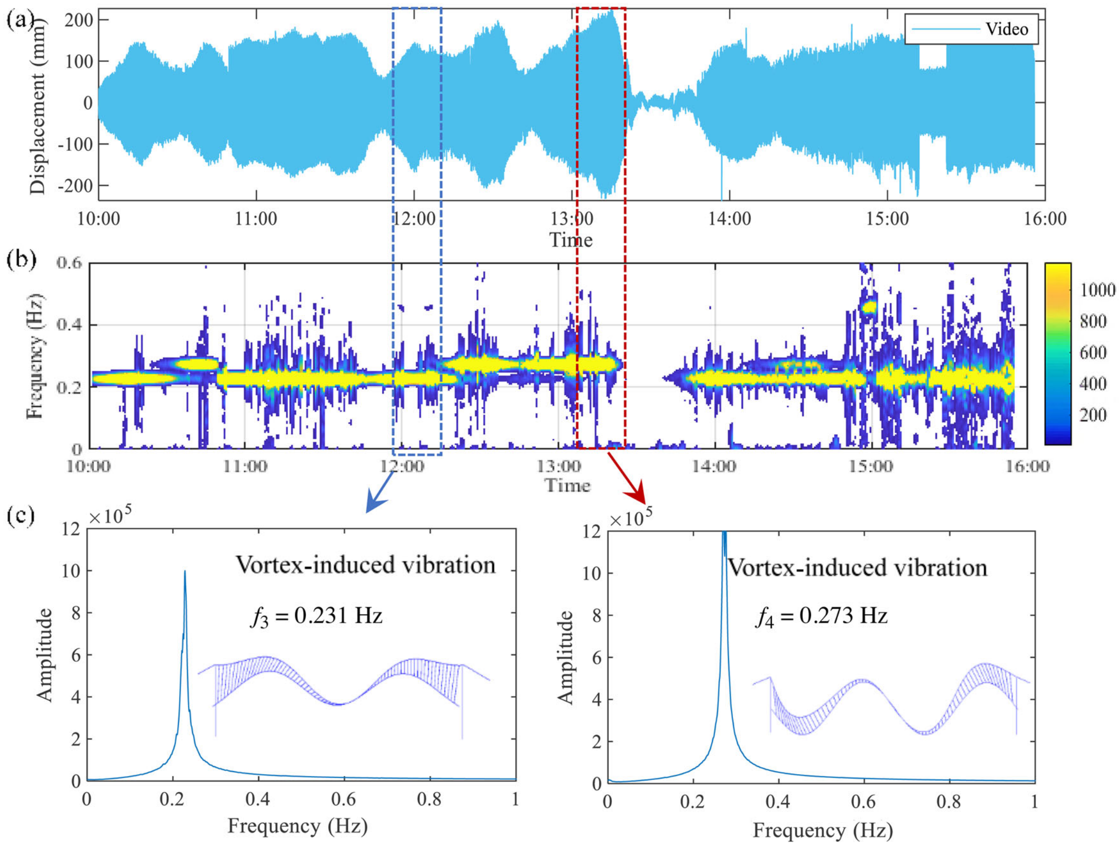

FIGURE 8 Vibration mode switching identified from video-captured displacements: (a) extracted displacements from video device; (b) time-frequency spectrum; (c) Fourier spectrum of corresponding displacements

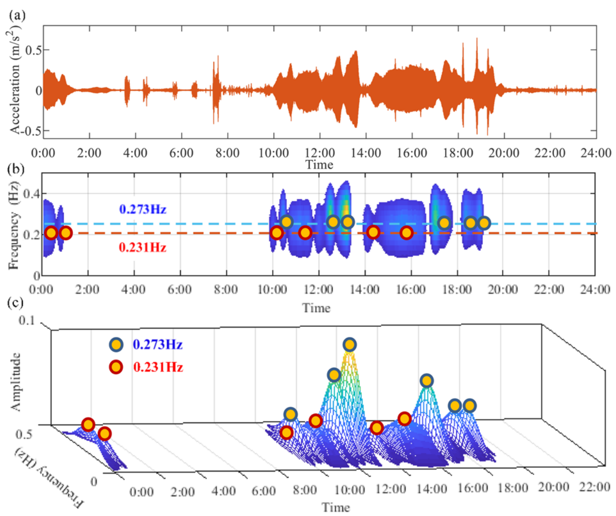

FIGURE 9 Vibration mode switch identified from measured accelerations: (a) dynamic acceleration at A1 location on May 9; (b) two-dimensional diagram of time-frequency spectrum; (c) three-dimensional diagram of time-frequency spectrum

the vibration mode switch phenomenon of the VIV event. It shows that not only the frequency component is switching, the energy amplitude is also changing with time.

In the above analysis for investigating the phenomenon of vibration mode switch, only dynamic responses at non-modal node (denoted as A1) are employed for analysis. The effect of modal nodes (referred to zero values of a specific vibration mode shape) on the identification of the VIV-triggered mode is investigated in this section. For this studied bridge, the VIV-triggered mode is occurred on the second, third and fourth mode, and dynamic responses measured at A2, A4, and A6 are located at the modal nodes of the fourth mode as shown in Figure 10a. The VIV-triggered mode is recognized as happening at the third mode (0.231 Hz) if accelerations measured at modal nodes (A2, A4, and A6) are adopted for Fourier spectrum analysis. This is due to the reason that modal responses corresponding to the fourth mode cannot be measured, so the VIV-triggered mode is identified incorrectly. On the other hand, dynamic responses at non-modal nodes (A1, A3, A5, and A7) can be adopted for further analysis, because the measured responses contain all frequency components of the bridge. The dominant frequency of 0.273 Hz is identified from the Fourier spectrum when the acceleration measured at A1 and A3 locations are used, whereas the dominant frequency is identified as 0.231 Hz when acceleration responses at A2 and A4 nodes are analyzed

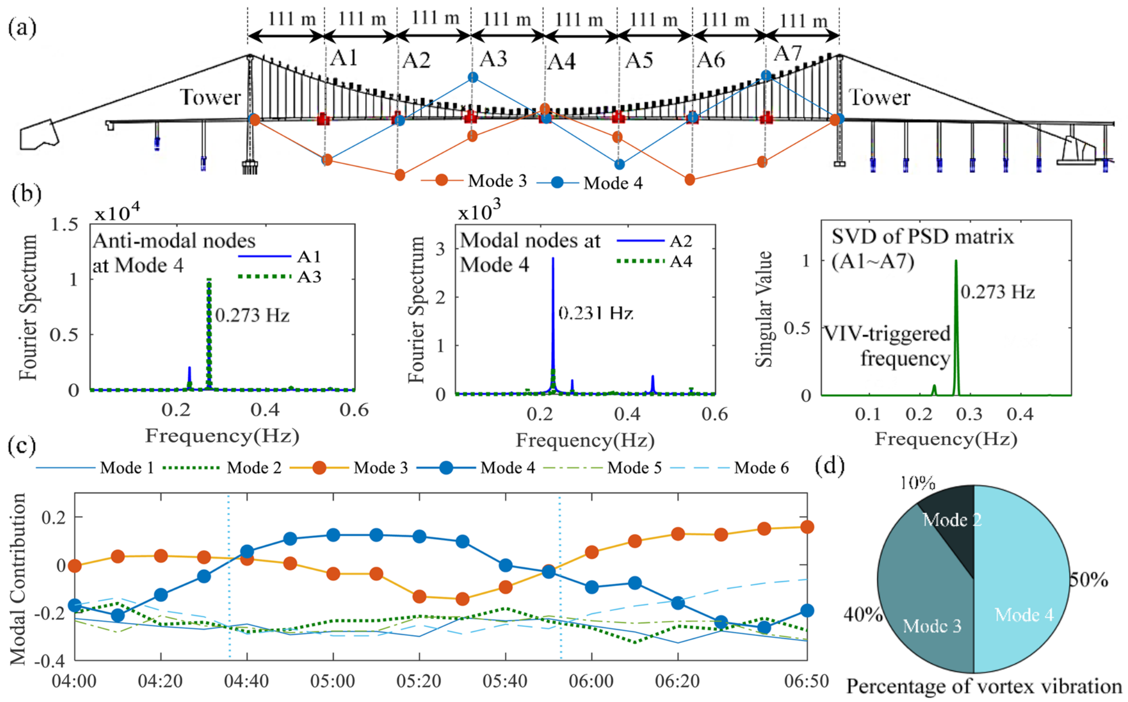

FIGURE 10 The effect of measurement locations on the identification of vortex-induced vibration (VIV)-triggered vibration mode: (a) schematic diagram of the third and fourth mode shape; (b) demonstration of the effect of modal nodes on VIV-triggered identification; (c) modal contribution; (d) percentage of VIV events from May 8 to May 15

(Figure 10b). Another difference is that the spectrum amplitude at 0.273 Hz is approximately 10 times than that at 0.231 Hz. To avoid this inconsistency, the SVD of power spectral density (PSD) matrix of acceleration responses at seven locations is adopted to identify the VIV-triggered mode. The VIV-triggered mode is recognized as happening at the fourth mode (0.273 Hz) from the SVD spectrum. In addition, the acceleration data at A1–A7 locations are divided into several data segments for a time period of 10 minutes to investigate the modal contribution of each mode when VIV event occurs. As shown in Figure 10c, it is found that the third and fourth vibration mode constitutes a large proportion during this period and are switching with each other, further demonstrating the phenomenon of vibration mode switch. According to analyzing measured data from May 8 to May 15, the total number of VIV events is found to be 90 times, by separating the VIV event corresponding to different vibration modes by investigating the characteristics of SVD spectrum, the percentage of VIV event happens at the second, third, and fourth mode are 10%, 50%, and 40%, respectively (Figure 10d).

4.3 | Relationship between dynamic response and wind conditions

The relationship between measured dynamic responses and wind conditions during VIV event will be investigated in this section, which is helpful for understanding wind characteristics of the bridge when VIV event occurs. The measured data from May 8 to May 26 are adopted to investigate the statistical relationship between the wind conditions and measured responses as shown in Figure 11. According to the above analysis, the occurrence of VIV event is accurately identified and then the measured data are divided into four groups, namely, VIV-trigged vibration at the second, third and fourth mode and normal vibrations. Each data segment with time duration of 10 minutes was employed for analysis. It can be seen from Figure 11a that the wind environment of the studied bridge under VIV event is characterized by southeast wind direction. According to the statistical relationship between RMS of acceleration/displacement responses and wind conditions (Figure 11b and c), we can see that the RMS value of vibration responses of the bridge under VIV event is much larger than that of normal vibration, which is the same as the conclusion from preliminary results. Another finding is that different VIV-triggered vibration has different wind speeds but with almost the same wind direction. The wind speed and direction around the bridge when VIV event occurs are in the range of 5–9 m/s and 290–330 degrees.

4.4 | Early warning framework for VIV events

A framework for early warning of VIV event is proposed in this section and this framework will be applied to the monitoring data of the studied bridge by incorporating

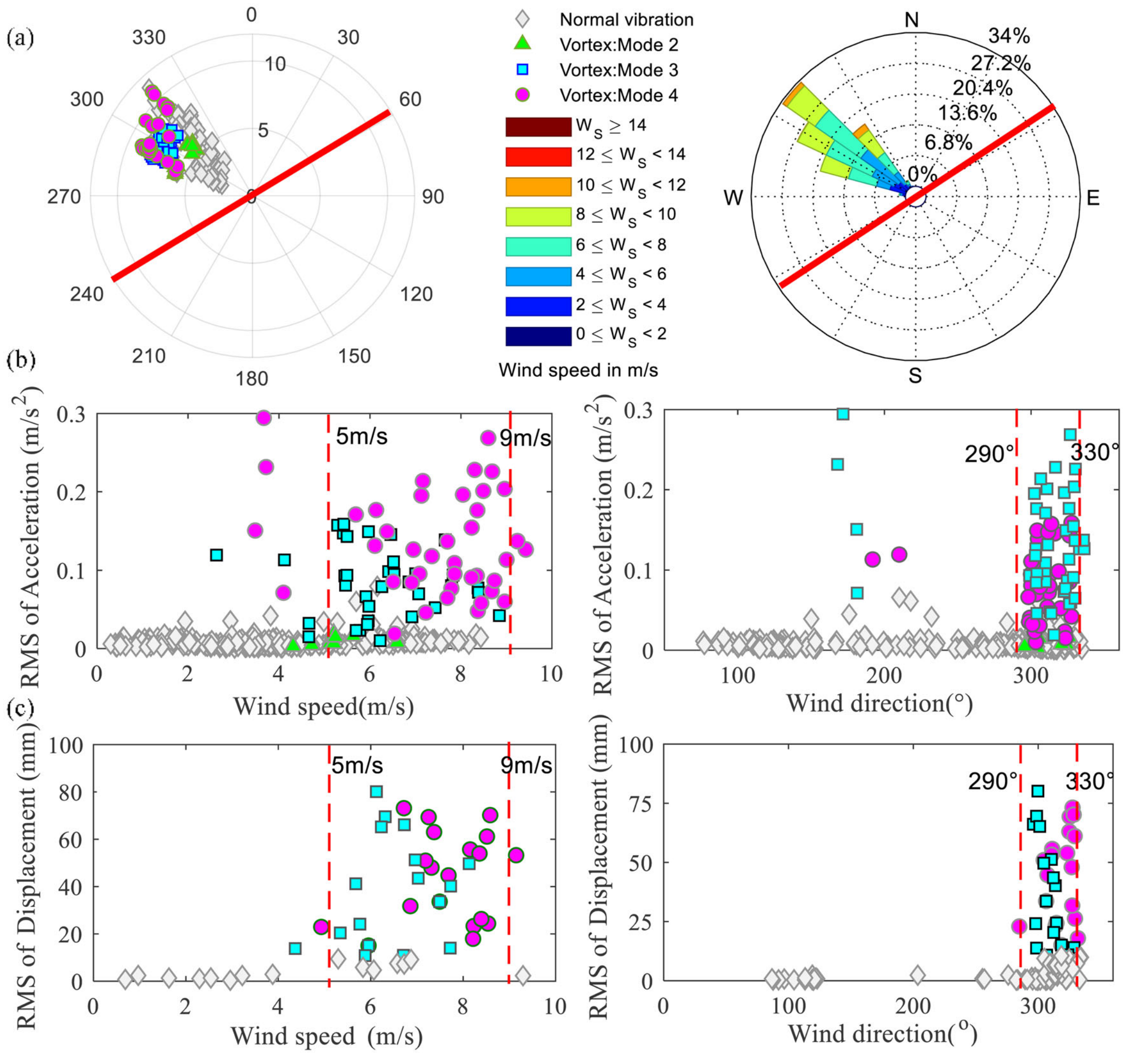

FIGURE 11 Statistical relationship between vortex-induced vibration (VIV) and wind field: (a) wind conditions; (b) the relationship between measured acceleration and wind fields; (c) the relationship between measured displacement and wind fields

identified characteristics of VIV events. The effectiveness of the proposed framework was successfully verified by using measured data at different periods of this bridge, the occurrence of VIV event can be successfully identified by using our proposed framework and warning indicators.

The proposed early warning framework for identifying VIV event is shown in Figure 12. The monitoring data including dynamic displacements, accelerations, and wind fields are employed to establish early warning indicators. In this framework, the bridge response and wind conditions data in every 10 minutes segment are used to analyze the bridge state and cycle forward in time, the single-mode characteristics of VIV event is taken as the most important factor to distinguish VIV signals from measured data, and the RMS value of dynamic responses and wind fields are used as secondary indicators to determine whether there is a need to trigger early warning system or not, the parallel condition is used to judge the vortex induced vibration, rather than the serial condition. Specifically, measured dynamic responses (i.e., displacements and accelerations) of the studied bridge are processed in frequency domain and time domain, respectively, and all thresholds are defined according to our statistical results. In frequency domain, natural frequencies and mode shapes are obtained by using modal analysis algorithms, then the energy concentration factor (ECF) Q_E (range : 0–1) can be calculated by using the proportion of the highest peak $p_{\max}$ in the SVD spectrum of PSD matrix of responses at all locations, which shows that if VIV occurs, the energy concentration factor value is close to 1, and the threshold of ECF is set as $Q'_E = 0.9$. In the time domain, bridge shape at a specific time instant can be obtained from monitored multiple-point displacements, which is further adopted to calculate similarity index with previously identified mode shapes as an early warning indicator (MAC value: Q_M), which is proposed by our team for the first time, and its basic principle

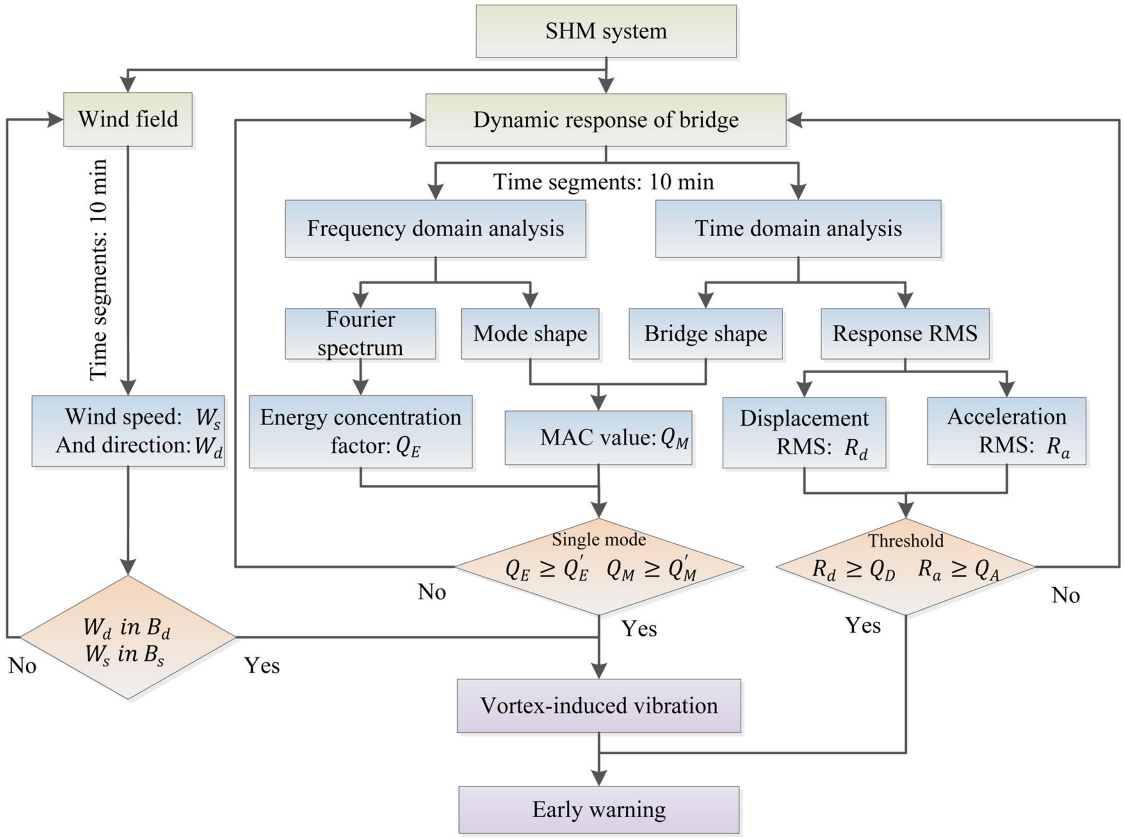

FIGURE 12 Proposed framework for early warning of vortex-induced vibration (VIV) event

has been explained in Section 4.1, that is, if VIV is in the form of single-mode vibration, the vibration form should also be dominated by the corresponding mode vibration mode, as shown in Figure 8, and the threshold of the MAC value is set as $Q'_M = 0.9$. Therefore, we calculate the similarity between the bridge type and the vibration mode of the bridge and adopt the confidence level consistent with the energy concentration coefficient. Once the above two indexes are satisfied, we know that the bridge is vibrating in a single frequency. The wind characteristics of the studied bridge during VIV are statistically and explained in Section 3 of this paper. If the wind fields are monitored and the average wind speed (W_s) is in the range of B_s : 5–9 m/s and the wind direction (W_d) is in the range of B_d : 290–330 degrees, it can be determined that a VIV event is occurring on the bridge. We need to further determine the RMS value of measured responses to judge the necessity of triggering early warning system by using the RMS value of measured responses. The RMS value of displacements and accelerations are set according to the statistical results in Figure 12, the statistical results of RMS based on acceleration show that when VIV occurs, the RMS value of the bridge will exceed 0.05 m/s^2. The statistical results of displacement show that the RMS value of the bridge will exceed 20 mm when VIV occurs. The threshold values of displacement and acceleration RMS are set as $Q_D = 20$ mm and $Q_A = 0.05$ m/s^2, respectively, according to bridge specifications. If both the displacement RMS (R_d) and acceleration RMS (R_a) exceed the defined threshold, the early warning system needs to be triggered immediately.

The proposed framework is applied to the studied bridge for identifying VIV event. The data processing flowchart and results are plotted in Figure 13. Measured multiple point displacements of the bridge on May 8 (VIV event) are employed to demonstrate the effectiveness and reliability of the proposed framework. The SVD is firstly performed on the PSD matrix of monitored dynamic displacement to obtain the singular value spectrum. The results show that the singular value spectrum of measured displacement on May 13 has a unique peak around 0.273 Hz, and the similarity between the bridge shape obtained at several time instants and the corresponding vibration mode shape at 0.273 Hz is larger than 98%, demonstrating the bridge exhibiting single-mode characteristics at the fourth mode (0.273 Hz). Measured data at non-modal nodes (A1, A3, A5, A7) corresponding to the fourth mode shape are selected for further calculating the RMS value of monitored responses. It can be seen that the RMS value of measured displacement and accelerations are exceeding the defined threshold value, indicating the warning system

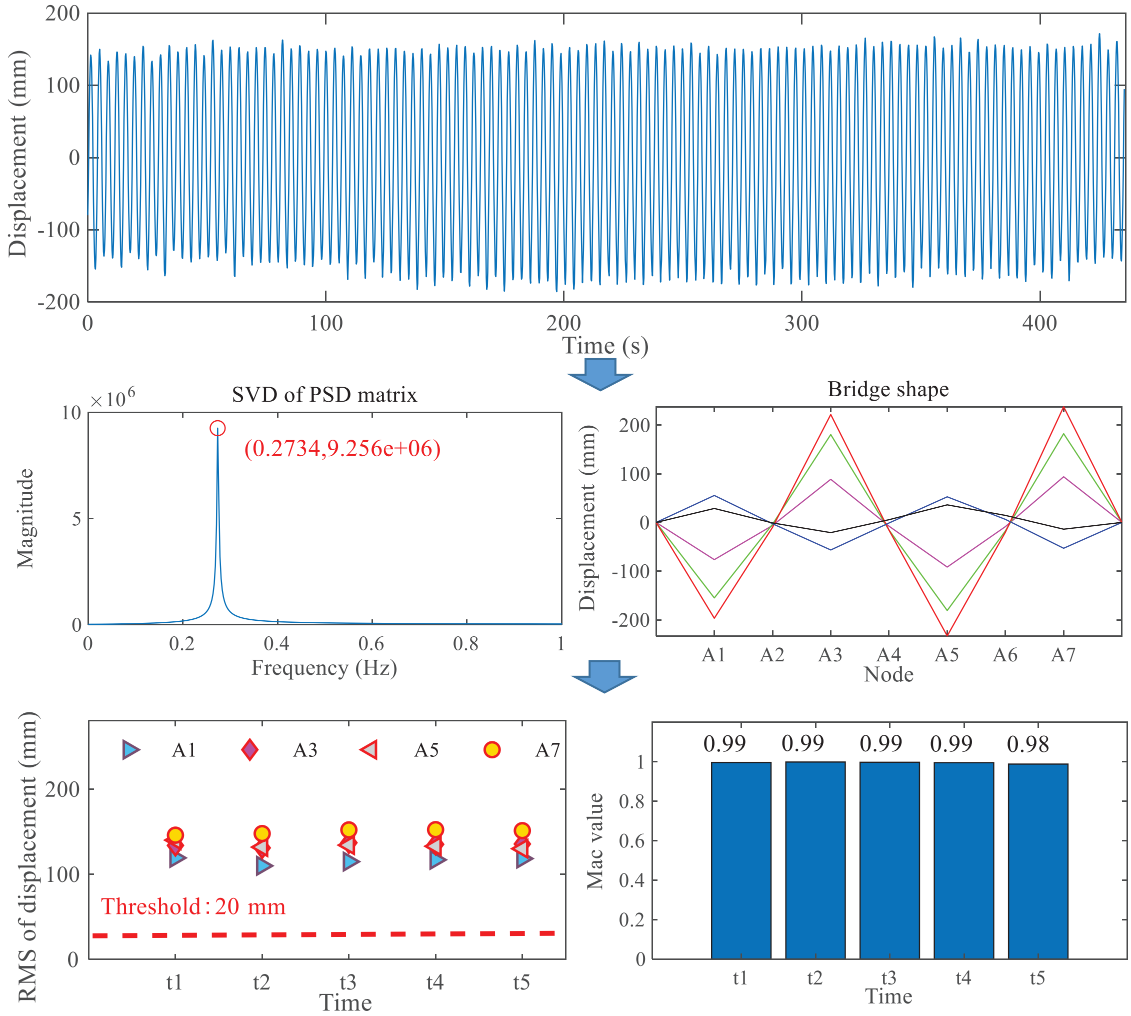

FIGURE 13 Vortex-induced vibration (VIV) warning of the studied bridge by using proposed framework

needs to be trigged for taking vibration mitigation measures. In a nutshell, if the measured data have the characteristics of single-mode vibration and the vibration amplitude exceeds the defined threshold, there is a need to trigger the warning system and take some mitigation measures (i.e., installing TMD dampers).

5 | CONCLUSIONS

Long-span bridges are susceptible to VIV event at low wind speeds because of the lower and denser natural frequencies. The VIV event usually has a large vibration amplitude and makes drivers/pedestrians feel uncomfortable, and even lead to fatigue damage if the VIV event lasts for a long time. Noncontact sensing strategies were proposed in this article to measure the VIV and a framework was presented for early warning of long-span bridges. Specific conclusions are drawn as follows:

Noncontact sensing technologies including microwave radar, optical camera, and video device were employed to monitor multiple point dynamic displacements of a long-span suspension bridge when VIV event occurs. The obtained results from three noncontact sensing devices were compared with each other to verify the correctness of measured displacements.

Dynamic properties including natural frequencies, damping ratios and modal shapes of the bridge were identified by using monitored data.

Singular value spectrum of multiple-point responses of the studied bridge was adopted to determine the dominant frequency components to avoid the influence of measured responses at modal nodes on the characteristic of Fourier spectrum, the VIV-triggered vibration of the studied bridge occurs at the second (0.171 Hz), third (0.231 Hz), and fourth mode (0.273 Hz). The vibration mode of the bridge is switching between the third and fourth vibration mode. The results show that the VIV event has the characteristics of single-mode vibration and vibration mode switching.

The similarity between the bridge shape at a specific time instant and the dominant vibration mode shape was proposed for further confirming the single-mode vibration characteristics of a VIV event. The results show that the correlation coefficient between bridge shape and the fourth vibration mode shape exceeds 0.9, indicating the occurrence of VIV event.

The proposed noncontact sensing technologies and early warning framework will be helpful for the identification and warning of VIV event of long-span bridges and provide some insights to the bridge owners and government for taking vortex mitigation measures.

ACKNOWLEDGMENTS

The research presented was financially supported by the National Key R&D Program of China (2018YFC0705601), National Science Foundation of China (51778134), Research project of Guangdong Provincial Communication Group Company Limited (2020103.1), Key R&D Project of Hebei Province (19210804D) and Science and Technology Innovation Project of China Energy Investment Corporation (SHGF-15-41).

REFERENCES

Amies, A. C., Pretty, C. G., Rodgers, G. W., & Chase, J. G. (2019). Experimental validation of a radar-based structural health monitoring system. *IEEE/ASME Transactions on Mechatronics*, *24*(5), 2064–2072.

Baker, S., & Matthews, I. (2004). Lucas-Kanade 20 years on: A unifying framework. *International Journal of Computer Vision*, *56*(3), 221–255.

Brownjohn, J. M. W., Stefano, A. D., Xu, Y. L., Wenzel, H., & Aktan, A. E. (2011). Vibration-based monitoring of civil infrastructure: Challenges and successes. *Journal of Civil Structural Health Monitoring*, *1*(3–4), 79–95.

Busca, G., Cigada, A., Mazzoleni, P., & Zappa, E. (2014). Vibration monitoring of multiple bridge points by means of a unique vision-based measuring system. *Experimental Mechanics*, *54*(2), 255–271.

Catbas, F. N., Brown, D. L., & Aktan, A. E. (2004). Parameter estimation for multiple-input multiple-output modal analysis of large structures. *Journal of Engineering Mechanics*, *130*(8), 921–930.

Chen, Z. S., Zhou, X., Wang, X., Dong, L. L., & Qian, Y. H. (2017). Deployment of a smart structural health monitoring system for long-span arch bridges: A review and a case study. *Sensors*, *17*(9), 2151.

Frandsen, J. B. (2001). Simultaneous pressures and accelerations measured full-scale on the great belt east suspension bridge. *Journal of Wind Engineering & Industrial Aerodynamics*, *89*(1), 95–129.

Fujino, Y., & Yoshida, Y. (2002). Wind-induced vibration and control of Trans-Tokyo Bay Crossing Bridge. *Journal of Structural Engineering*, *128*(8), 1012–1025.

Hao, J. M., & Wu, T. (2018). Downburst-induced transient response of a long-span bridge: A CFD-CSD-based hybrid approach. *Journal of Wind Engineering and Industrial Aerodynamics*, *179*, 273–286.

Hwang, Y. C., Kim, S., & Kim, H. K. (2019). Cause investigation of high-mode vortex-induced vibration in a long-span suspension bridge. *Structure and Infrastructure Engineering*, *16*(8), 1–10.

Kim, H., & Adeli, H. (2005). Wind-induced motion control of 76-story benchmark building using the hybrid damper-tuned liquid column damper system. *Journal of Structural Engineering*, *131*(12), 1794–1802.

Li, H., Laima, S., Zhang, Q., Li, N., & Liu, Z. (2014). Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *Journal of Wind Engineering & Industrial Aerodynamics*, *124*, 54–67.

Li, H., Laima, S., Ou, J., Zhao, X., Zhou, W., Yu, Y., Li, N.a., & Liu, Z. (2011). Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Engineering Structures*, *33*(6), 1894–1907.

Lynch, J. P. (2007). An overview of wireless structural health monitoring for civil structures. *Philosophical Transactions Mathematical Physical & Engineering Sciences*, *365*, 345–372.

Macdonald, J. H. G., Irwin, P. A., & Fletcher, M. S. (2003). Vortex-induced vibrations of the Second Severn Crossing cable-stayed bridge: Full-scale and wind tunnel measurements. *Structures & Buildings*, *152*(2), 123–134.

Nasslf, H. H., Gindy, M., & Davis, J. (2005). Comparison of laser doppler vibrometer with contact sensors for monitoring bridge deflection and vibration. *NDT & E International*, *38*(3), 213–218.

Nagai, M., Fujino, Y., Yamaguchi, H., & Iwasaki, E. (2004). Feasibility of a 1,400 m span steel cable-stayed bridge. *Journal of Bridge Engineering*, *9*(5), 444–452.

Ni, F. T., Zhang, J., & Noorimohammad, N. (2019). Deep learning for data anomaly detection and data compression of a long-span suspension bridge. *Computer-Aided Civil and Infrastructure Engineering*, *35*(7), 685–700.

Ni, F. T., Zhang, J., & Chen, Z. Q. (2019). Pixel-level crack delineation in images with convolutional feature fusion. *Structural Control Health Monitoring*, *26*, e2286.

Ou, J. P., & Li, H. (2010). Structural health monitoring in mainland China: Review and future trends. *Structural Health Monitoring*, *9*(3), 219–231.

Oh, B. K., Kim, K. J., Kim, Y., Park, H. S., & Adeli, H. (2017). Evolutionary learning based sustainable strain sensing model for structural health monitoring of high-rise buildings. *Applied Soft Computing*, *58*, 576–585.

Park, H. S., Lee, H. M., Adeli, H., & Lee, I. (2007). A New approach for health monitoring of structures: Terrestrial laser scanning. *Computer-Aided Civil and Infrastructure Engineering*, *22*(1), 19–30.

Park, S. W., Park, H. S., Kim, J. H., & Adeli, H. (2015). 3D displacement measurement model for health monitoring of structures using a motion capture system. *Measurement*, *59*, 352–362.

Pan, B., Tian, L., & Song, X. (2016). Real-time, non-contact and targetless measurement of vertical deflection of bridges using off-axis digital image correlation. *NDT & E International*, *79*, 73–80.

Pieraccini, M., Miccinesi, L., Nejad, A. A., & Fard, A. N. N. (2019). Experimental dynamic impact factor assessment of railway bridges through a radar interferometer. *Remote Sensing*, *11*(19), 2207

Powell, M. D., Vickery, P. J., & Reinhold, T. A. (2003). Reduced drag coefficient for high wind speeds in tropical cyclones. *Nature*, *422*(6929), 279–283.

Ribeiro, D., Calcada, R., Ferreira, J., & Martins, T. (2014). Non-contact measurement of the dynamic displacement of railway bridges using an advanced video-based system. *Engineering Structures*, *75*(Sep.15), 164–180.

Scanlan, R. H. (1981). *State-of-the-art methods for calculating futter, vortex-induced and buffeting response of bridge structures* (Federal Highway Administration Report, FHWA/RD-80/050), National Technical Information Service, USA.

Sohn, H., Farrar, C. R., Hemez, F. M., Shunk, D. D., Stinemates, D. W., Nadler, B. R. (2004). A Review of Structural Health Review of Structural Health Monitoring Literature 1996-2001. data acquisition.

Tao, T., Xu, Y. L., & Huang, Z. F. (2020). Buffeting analysis of long-span bridges under typhoon winds with time-varying spectra and coherences. *Journal of Structural Engineering*, *146*(12), 04020255.

Tian, Y. D., Zhang, C., Jiang, S., Zhang, J., & Duan, W. H. (2020). Non-contact cable force estimation with unmanned aerial vehicle and computer vision. *Computer-Aided Civil and Infrastructure Engineering*, *36*, 73–88. https://doi.org/10.1111/mice.12567

Wang, N., & Adeli, H. (2015). Robust vibration control of wind-excited highrise building structures. *Journal of Civil Engineering and Management*, *21*(8), 967–976.

Xiang, H. F., Ge, Y. J., Zhu, L. D., Chen, A. R., Lin, Z. X., Gu, M., & Xiao, R. C. (2005). *Modern theory and practice on bridge wind resistance*. China Communication Press.

Xu, Y., Brownjohn, J., & Kong, D. (2018). A non-contact vision-based system for multipoint displacement monitoring in a cable-stayed footbridge. *Structural Control & Health Monitoring*, *25*(5), e2155.

Xu, Y., & Chan, W. (2009). Wind and structural monitoring of long span cable supported bridges with GPS. The Seventh Asia-Pacific Conference on Wind Engineering. Taipei, Taiwan.

Yu, S., Zhang, J., & He, X. (2020). An advanced vision-based deformation measurement method and application on a long-span cable-stayed bridge. *Measurement Science & Technology*, *31*, 065201.

Yu, S., & Zhang, J. (2019). Fast bridge deflection monitoring through an improved feature tracing algorithm. *Computer-Aided Civil and Infrastructure Engineering*, *35*(3), 292–302.

Yoon, H., Shin, J., & Spencer, B. F. (2020). Structural displacement measurement using an unmanned aerial system. *Computer-Aided Civil and Infrastructure Engineering*, *33*(3), 183–192.

Zhang, G., Wu, Y., Zhao, W., & Zhang, J. (2020). Radar-based multipoint displacement measurements of a 1200-m-long suspension bridge. *ISPRS Journal of Photogrammetry and Remote Sensing*, *167*, 71–84.

Zhang, B., & Zhang, J. (2020). A traffic surveillance system for obtaining comprehensive information of the passing vehicles based on instance segmentation. *IEEE Transactions on Intelligent Transportation Systems*, *pp*(99), 1–16. https://doi.org/10.1109/TITS.2020.3001154.

Zhao, W., Zhang, G., & Zhang, J. (2020). Cable force estimation of a long-pan cable-stayed bridge with microwave interferometric radar. *Computer-Aided Civil and Infrastructure Engineering*, *35*(12), 1419–1433.

Zhou, Y., & Sun, L. (2018). Effects of high winds on a long-span sea-crossing bridge based on structural health monitoring. *Journal of Wind Engineering and Industrial Aerodynamics*, *174*, 260–268.

Zhu, L. D., Gao, G. Z., & Zhu, Q. (2020). Recent advances, future application and challenges in nonlinear flutter theory of long span bridges. *Journal of Wind Engineering and Industrial Aerodynamics*, *206*, 104307.

Zhu, L., Meng, X., & Guo, Z. (2013). Nonlinear mathematical model of vortex-induced vertical force on a flat closed-box bridge deck. *Journal of Wind Engineering and Industrial Aerodynamics*, *122*, 69–82.

How to cite this article: Zhang J, Zhou L, Tian Y, Yu S, Zhao W, & Cheng Y. (2022). Vortex-induced vibration measurement of a long-span suspension bridge through noncontact sensing strategies. *Comput Aided Civ Inf*, *37,* 1617–1633. https://doi.org/10.1111/mice.12712